MILITARY SPECIFICATION

RADAR SET AN/APN-117

This specification has been approved by
the Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall utilize frequency modulation principles for determining and indicating the height (surface or terrain clearance) of an aircraft from 0 to 1000 feet, and for supplying control signals to an automatic pilot system and other devices when the equipment is properly installed in the aircraft.

1.2 Classification - The radar set shall be of one type and shall consist of the following:

Units	Type Designation	Max. Allow. Size (Inches) W	H	L/D	Max. Allow. Wt. Lbs	Appl. Para.
Receiver-Transmitter	RT-160/APN-22	3-49/64	4-1/16	12-17/64	4.8	3.5.1
Housing, Receiver-Transmitter	CY-1409A/APN-22	5-45/64	4-3/16	14-5/8	2.25	3.5.1.2
Amplifier, Electronic Control	AM-1717/APN-117	5-1/2	7-61/64	22-29/32	19.5	3.5.2
Mounting Base	MT-803/APN-22				4.0	3.5.2.2
Control-Indicator, Height	C-2236/APN-117	3-9/16	3-17/64	6-1/16	2.0	3.5.3

1.3 Accessories - The equipment shall be capable of operating with the following accessories which are not to be supplied as part of this equipment.

Automatic Pilot System

Deviation Indicator

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-Z-285 Zinc: Anodes

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Military

MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For Use with Electronic Equipment in Aircraft)
MIL-E-4682	Electron Tubes, Choice and Application of
MIL-C-5015	Connectors; Electrical, "AN" Type
MIL-L-5057	Light, Panel Aircraft, Individual Instrument
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Limits, Tests, and Design Requirements; Aircraft Electrical and Electronic Equipment
MIL-E-7894	Electric Power, Aircraft, Characteristics of
MIL-S-16892	Synchros, 400 Cycles
MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment; Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment

STANDARDS

Federal

FED-STD-595	Colors

DRAWINGS

Air Force - Navy Aeronautical

MS 33558	Numerals and Letters; Aircraft Instrument Dial, Standard Form of
AND 10402	Aircraft Instrument Case

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PUBLICATIONS

Bureau of Aeronautics

SAR-300 — Engineering Change Proposals, Electronic Material Changes and Electronic Material Bulletins; Preparation of

2.2 Availability of Documents:

(1) Give both the title and number when requesting specifications, standards, drawings and publications. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania. Attention: Code ODPT.

(3) Copies of applicable drawings may be obtained upon application to the Bureau of Aeronautics, Navy Department, Washington 25, D.C., Attention: Avionics Division.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring agency.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the characteristics of standard MIL parts are such that their use will not fulfill the requirements because of size, weight, performance or other reasons, material and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specifications.

(3) When a non-standard part is used, the meeting of general material requirements such as non-inflammability, fungus resistance, non-toxicity, etc., and, the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall be approved first by the Bureau of Aeronautics.

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3.1.1 Non-Standard Part and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400, except as follows:

(1) Mechanical parts are expected to meet the material requirements of Specification MIL-E-5400, and shall be submitted as non-standard parts only when the material requirements are not met.

(2) Non-standard parts, as defined in Specification MIL-E-5400 of an electrical nature shall be approved before the preproduction model of the equipment is delivered. The request for approval shall be forwarded by the contractor before the part is purchased or fabricated. In addition to or in lieu of the Non-Standard Part Data Sheet, the contractor is requested to submit itemized lists of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.1.2 Choice of Tube Types - Electron tubes used in the equipment shall be chosen in accordance with the following requirements and in the sequence noted:

(1) MIL-E-4682

(2) The use of any tube not meeting the requirements of (1) above shall be approved by the Bureau of Aeronautics; however, before consideration for approval, the contractor shall submit the following information:

(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(c) Information as to whether the tube manufacturer has received Military or Navy qualification approval for the particular tube.

3.1.2.1 Electron Tube Complement - A tube complement report shall be prepared in accordance with Specification MIL-E-4682 and submitted to the Bureau of Aeronautics for review and approval prior to preproduction testing. The tube complement shall include all electron tube types (including crystal diodes and transistors) used in the equipment.

3.1.3 Plug-In Assemblies - The use of easily disconnected plug-in assemblies of one or more related circuits shall be followed, to reduce size and weight, to simplify operation and maintenance, or to improve performance.

3.1.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request shall be accompanied by complete supporting information.

3.2 Design and Construction:

3.2.1 General - This equipment shall fulfill all the applicable design and construction requirements as specified in Specification MIL-E-5400, except as otherwise specified herein.

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3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment.

3.2.3 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment:

Temperature	Room ambient 30° ± 10°C
Altitude	Normal ground
Vibration	None (Except indicator)
Humidity	Room ambient up to 90% relative humidity
Input Power	115 ± 1.0 V AC and/or 27.5 ± 0.5 V DC

3.2.4 Orientation - The equipment shall operate in one position with respect to the vertical, but shall also operate normally and reliably in an aircraft when climbing and diving at 50° angles and when banking at 30° angles, taking into consideration the effects of the antenna pattern specified herein. No detrimental effects shall be caused to the operation of the equipment by any flight maneuver, and operation of the equipment in any position shall not prevent normal operation when the aircraft returns to normal flight.

3.2.5 Rate of Climb and Dive - The rate of vertical ascent may be as great as 200 feet per second and may be experienced at any altitude above 200 feet. The vertical component of velocity during dives may reach, but will not exceed $\frac{600A}{3000}$ feet per second where A is the height of the aircraft above the terrain in feet.

3.2.6 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 32.6 pounds.

3.2.7 Operational Stability - The equipment shall operate with optimum performance for 100 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.8 Operating Life

3.2.8.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 300 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.8.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.9 Interconnection Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded, when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

3.2.10 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with the requirements of Specification MIL-E-5400. However, cables and that portion of the connectors which is to be attached to the cables shall not be supplied as part of the equipment.

3.2.11 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in Specification MIL-E-7894 having limits as specified herein. The power required shall not exceed the specified amounts.

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(1) AC Power (Single Phase)

(a) 115-Volt Variable Frequency

Operating Voltage Limits	108 to 121 volts
Operating Frequency Limits	320 to 1000 cps
Reduced Performance Voltage Limits	102 to 108 and 121 to 124 volts
Power (1Ø)	130 V. A.

(b) 115-Volt Constant Frequency

Operating Voltage Limits	108 to 121 volts
Operating Frequency Limits	380 to 420 cps
Reduced Performance Voltage Limits	102 to 108 and 121 to 124 volts
Power (1Ø)	25 V. A.

(2) DC Power 28-Volt System

Operating Voltage Limits	25. 0 to 29. 0 volts
Reduced Performance Voltage Limits	20 to 25 and 28 to 29 volts
Current	1. 5 amps

3. 2. 11. 1 Grounded Input - The equipment shall be such that the following power "lines" may be grounded externally:

(1) One side of the single-phase power sources

(2) The negative of the d. c. power source.

3. 2. 12 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3. 2. 12. 1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3. 2. 12. 2 Input Power Connections - Input power connections shall be made to the suitable contacts in the dual female receptacle attached to the Mounting Base MT-803/APN-22.

3. 2. 13 Moisture and Fungus Resistant Treatment - Equipments shall be fungus proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary.

3. 2. 14 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307.

3. 2. 15 Interchangeability - The interchangeability requirements between like assemblies, subassemblies and replaceable parts as specified in Specification MIL-E-5400 shall be extended to require interchangeability of units of this equipment with the latest model of any previously produced equipment. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Bureau of Aeronautics for the purpose. In the event that requirements of Specification MIL-E-5400 conflict with the requirements for interchangeability, the interchangeability requirements shall govern.

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3.2.16 Provisions for Maintenance and Field Testing - The equipment shall provide for ease of maintenance to the greatest extent practicable. Provisions for field testing shall be as specified in Specification MIL-E-5400. The equipment shall provide test points connected to appropriate circuit functions, and terminated in the dual receptacle at the rear of the Electronic Control Amplifier AM-1717 APN-117.

3.2.17 Conducted Noise - The conducted noise limits specified in Specification MIL-I-6181 shall not apply to interconnecting leads between RT-160/APN-22 and AM-1717/APN-117, provided that such leads are not common to other equipment in a using aircraft.

3.3 Performance

3.3.1 General - When completely assembled in an operating system, the equipment shall meet the requirements of this specification.

3.3.2 Operation - The equipment when properly interconnected, shall perform the following functions:

(1) Continuously indicate the height (surface or terrain clearance) of the aircraft from 0 to 1000 feet.

(2) Indicate when the aircraft is flying at or below a prescribed height (surface or terrain clearance).

(3) Indicate when the equipment is not operating properly.

(4) Supply height control signals to the automatic pilot system and/or other equipment.

3.3.3 Accuracy - The overall system accuracy shall be not less than ± 5% or ± 2 feet of the correct terrain clearance whichever is the greater.

3.3.4 Frequency - The equipment shall operate within the frequency range of 4200 to 4400 megacycles with guard bands at both ends of the range to preclude interference with equipment operating outside of this range.

3.3.5 Modulation Characteristics - The signal shall be frequency-modulated at a rate sufficient to meet the accuracy requirements of this specification. The modulation shall be of the triangular type, that is, the frequency versus time characteristics shall be essentially linear throughout each half sweep and the slope during the downsweep shall be essentially the negative of the slope during the upsweep.

3.3.6 Transmitter Power Output - The transmitter shall employ a type 6177 magnetron or equivalent having a minimum power output of 1.0 watt. The magnetron shall look into a connecting cable load having a maximum connector mismatch of 1.5 VSWR and a maximum attenuation of 1.0 db.

3.3.7 Receiver Gain - The receiver shall have a gain characteristic such that optimum signal output level shall be provided at all times.

3.3.8 Receiver Sensitivity - The sensitivity of the receiver-amplifier shall be such that an audio signal of not more than 2 microvolts applied to its input is required for stable limiting at the peak frequency of the receiver-amplifier.

3.3.9 Balanced Detector Performance - With the equipment properly adjusted and the detector properly balanced using paired 1N21C crystals, the equipment shall discriminate against unwanted transmitter amplitude modulation within the frequency range by a factor of not less than 30 db when transmitting into free space.

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3.3.10 Blocking - The receiver shall be such that a variation in signal level of 80 db from sweep to sweep does not cause blocking, overload or otherwise affect the accuracy of the equipment by varying the average value of the height signal in the limiter.

3.3.11 Limiting - Limiting circuits shall substantially reduce noise and amplitude modulation and clip both the positive and negative peaks of the height signal symmetrically about its average value.

3.3.12 Tracking Speed - The equipment shall follow a maximum rate of change of altitude, in accordance with the formula 600A/3000 feet/second of indicated height (where A equals the altitude in feet).

3.3.13 Lockout - In the absence of an altitude signal, the equipment shall continuously scan throughout the entire altitude range without locking on any false altitude indication.

3.3.14 Warm-Up Period - The equipment shall require not more than 3 minutes after the power is turned on to become operative (indicate height to within 10% of the correct terrain clearance) and shall not require more than 12 minutes to meet the accuracy requirements of this specification; except that at temperatures below -40°C, the equipment shall not require more than 10 minutes to become operative (indicate height to within 10% of true height) and shall not require more than 25 minutes to meet the accuracy requirements of this specification. This requirement shall apply to service test conditions and standard test conditions.

3.3.15 Height Limit Indication - As the aircraft descends, the red warning signal of the height limit indicator shall be illuminated when the indicator pointer corresponds to the setting of the limit control marker, within ± 3°, and shall remain illuminated as long as the pointer indicates an altitude below the limit control marker setting. As the aircraft ascends, the red warning signal shall be extinguished when the indicator pointer passes the limit control marker, within ± 3°.

3.3.16 Reliability Indication - The control-height indicator pointer shall move off the calibrated portion of the dial when the signal strength becomes inadeqaute to provide reliable altitude readings. When the control-height indicator pointer moves off the calibrated portion of the dial, the coil voltage for an external relay in the autopilot circuitry shall be removed.

3.3.17 Height Control Signals - The equipment shall supply a control signal dependent upon the deviation of the surface or terrain clearance indication from the preselected height. These control signals shall be of such character as to be proportional to the deviation from the preselected altitude and may consist of two potentiometers connected as a bridge circuit, with one potentiometer actuated from the servo motor, and the other from the preselected height indicator control. Any external equipment will supply the form of control signal required across the potentiometers, and the difference in the positions of the two potentiometers shall supply the output control signal to the external equipment.

3.3.18 Residual Height Adjustment - The equipment shall be adjustable to allow for installation of the antenna in the aircraft from 2 to 20 feet above the ground.

3.4 Workmanship - Workmanship, fabrication and manufacturing processes shall conform to Specification MIL-E-5400.

3.5 Detailed Requirements

3.5.1 Receiver-Transmitter RT-160 APN-22 - This unit shall meet the following requirements:

3.5.1.1 Function - The receiver-transmitter shall transmit a radio frequency carrier which is frequency modulated at a rate and bandwidth sufficient to meet the performance requirements of this specification, and receive and combine the reflected signal with a signal obtained directly from the transmitter to produce an audio-frequency signal necessary for operation of the equipment.

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3.5.1.2 Form Factor - The receiver-transmitter shall be 3-49/64 inches wide by 4-1 16 inches high by 12-17 64 inches long and shall be completely contained within a 5-45/64 by 14-5/8 by 4-3 '16 inch Receiver-Transmitter Housing CY-1409A/APN-22. The unit shall include a dual horn antenna system suitable for flush mounting directly into the frame of the aircraft.

3.5.1.3 Contents of Unit - The unit shall include an antenna, a frequency-modulated transmitter and a balanced detector.

3.5.1.3.1 Antenna - A transmitter and receiver horn antenna shall be provided with means for internally injecting part of the transmitter energy into the detector. The voltage standing wave ratio of the antenna shall not exceed 3.0 to 1 for the receiving antenna and 1.5 to 1 for the transmitter antenna; the feed through attenuation between horns shall be greater than 40 db over the frequency band of 4200 to 4400 mc/s. The radiation pattern, measured to the 3 db points shall be not less than 50° by 60°, the larger angle being in the plane of the major axis of the equipment.

3.5.1.3.2 Transmitter - The transmitter frequency modulated signal shall be generated by a frequency modulated continuous wave magnetron.

3.5.1.3.3 Balanced Detector - The balanced detector shall consist of a short-length of waveguide which contains two high-frequency crystal diodes connected to a special high-frequency circuit near one end and to a horn radiator at the other end.

3.5.1.4 Weight - The weight of this unit, including the Receiver-Transmitter Housing CY-1409A/APN-22, shall not exceed 7.1 pounds.

3.5.1.5 Electrical Connections - A terminal block mounted on the CY-1409A/APN-22 housing shall provide for all external electrical connections. Interconnection between RT-160/APN-22 and CY-1409A/APN-22 shall be accomplished by use of a suitable connector and mating receptacle. The magnetron socket and crystal connectors shall be of special design and in accordance with good engineering practice for this application.

3.5.1.6 Controls - There shall be no controls or adjustments located in the receiver-transmitter.

3.5.2 Electronic Control Amplifier AM-1717/APN-117 - This unit shall meet the following requirements:

3.5.2.1 Function - The electronic control amplifier shall utilize the surface or terrain clearance signals from the receiver-transmitter to operate and control the indicators and auxiliary output voltages and shall supply the necessary power conversion required by the equipment.

3.5.2.2 Form Factor and Mountings - The electronic control amplifier shall be completely housed in a case approved by the Bureau of Aeronautics, similar to the A1-D size specified in Specification MIL-C-172. Projection of the handles and other panel-mounted fixtures shall not extend more than 2 inches beyond the panel face. The unit shall be mounted on a Mounting Base MT-803/APN-22.

3.5.2.3 Contents of Unit - The unit shall contain a modulation source for the transmitter, a receiver amplifier, a servo amplifier, a servo mechanism controlled by altitude signals, a source of control voltages for the external indicators and accessory equipment, a reliability determining circuit, and power sources for the overall equipment.

3.5.2.3.1 Modulation Source - The function of the modulator shall be to drive the magnetron in such a manner that a symmetrical triangular frequency modulated output is obtained. Adjustments shall be provided for setting the bandwidth. When an altitude signal is not present, the modulation shall be varied over its deviation range in search of the altitude signals.

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3.5.2.3.2 Receiver Amplifier - The receiver amplifier shall employ an audio frequency amplifier, a limiter, a counter, and auxiliary circuits for determining the height by means of the reflected signal.

3.5.2.3.3 Servo Amplifier - This shall be a circuit which modifies and amplifies the receiver output signal to a form suitable for operating the servo motor.

3.5.2.3.4 Servo Mechanism - This unit shall contain motor driven potentiometers which, in combination, change the receiver bias potentials or vary the transmitter-modulation so that when the servo comes to rest, its position shall indicate the altitude on a servo indicator dial. A 115-volt, 400-cycle synchro in accordance with Specification MIL-S-16892 shall transmit its position to the indicators. During periods of inadequate signal, as determined by the reliability circuit, the servo will continuously scan the entire altitude range until a signal is again received. During the scan phase, the synchro transmitter shall be disconnected from the indicators. The control potentiometer for auxiliary equipment will be operated by this servo.

3.5.2.3.5 Reliability Circuit - The reliability circuit shall prevent the servo system from locking on noise, feed thru, microphonic or other interfering or false signal. When the altitude signal strength is insufficient to produce reliable height indications, the reliability circuit shall cause the indicator pointer to move off the calibrated portion of the dial, and shall cause the servo system to search for a true altitude signal.

3.5.2.4 Weight with Shock Mounts - The weight of this unit, including Mounting Base MT-803/APN-22, shall not exceed 23.5 pounds.

3.5.2.5 Electrical Connections - A dual multi-contact connector shall be mounted on the lower portion of the rear face of the amplifier chassis so as to automatically effect all power, control and output circuit connections to the mounting base when the amplifier is mounted on and attached thereto. An aperture shall be provided in the amplifier case to permit engagement of this connector with the mating connector on the mounting base. A sufficient quantity of terminals in the dual connector shall be provided for test purposes.

3.5.2.6 Controls - All controls necessary for adjusting and aligning the entire equipment, except for the servo gain control, sweep generator frequency control and operating controls, shall be located on the front panel.

3.5.3 Height Control-Indicator C-2236/APN-117 - This unit shall meet the following requirements:

3.5.3.1 Function - The control-height indicator shall indicate surface or terrain clearance by means of a single pointer against a fixed dial. A controllable marker reading against the same dial shall indicate the preselected height about which the control signals for the height limit indicator and other equipments operate. The height limit indicator shall warn of flight below the preselected height by means of a red light on the face of the instrument. The control-height indicator pointer shall move off the calibrated portion of the dial to indicate failure or unreliable operation of the equipment, when actuated by the reliability circuit.

3.5.3.2 Form Factor - The control-height indicator shall be completely housed in a case similar to that illustrated in Drawing AND 10402. The dial face shall be placed as closely as possible to the cover glass and the bezel shall be made as thin as possible in order not to obscure the dial markings when viewed at an angle. Maximum depth, excluding control knobs but including connector, shall not exceed 6-1/16 inches.

3.5.3.3 Contents of Unit - This unit shall include a 115-volt, 400-cycle synchro, in accordance with Specification MIL-S-16892, to rotate the altitude indicator pointer to a position determined by the servo-mechanism. It shall also include the height limit indicator and the height limit control marker and its mechanism.

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3.5.3.4 Weight - The weight of this unit shall not exceed 2.0 pounds.

3.5.3.5 Electrical Connection - A single multi-contact connector shall be mounted on the back of the indicator opposite the dial to provide for all electrical connections except illuminating wiring.

3.5.3.6 Controls - A control knob located at the corner of the AN type meter flange shall control the operation of the equipment, and function as the power switch, and the height limit selector. The height limit selector shall be continuously adjustable throughout the range of the equipment and shall control the error signal for the automatic pilot and deviation indicators, and the operation of the height limit marker and indicator.

3.5.3.7 Markings

3.5.3.7.1 Dial - The scale shall be divided into major and minor divisions to form a modified linear logarithmic type scale similar to that shown in figure 1. The corresponding servo indicator dial shall be as shown in figure 2.

3.5.3.7.2 Pointer - The pointer shall be as shown in figure 1.

3.5.3.7.3 Marker - The marker shall be as shown in figure 1.

3.5.3.8 Instrument Lights - One circuit of instrument lights shall be provided in a mask on the front of the housing for illuminating the dial face. The circuit shall contain lamps with red filters for illumination in accordance with Specification MIL-L-5057.

3.5.3.9 Height Limit Indicator - The height limit indicator shall consist of a manually controlled variable brilliance light which gives warning that the terrain clearance is below the preselected height. The lamp for the indicator shall be replaceable from the front of the indicator case without disturbing the installation. The light shall be grounded by a switch in the equipment, the other lead terminating in the connector at the back of the indicator.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine conformance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

(3) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the Bureau of Aeronautics.

4.2.2 Service Approval Tests - The service approval model shall consist of one equipment representative of the production equipment to be supplied under the contract. This may be the same equipment that was subjected to the contractor's demonstration tests. This equipment shall be delivered to the destination specified by the Bureau of Aeronautics. When the tests of 4.2.1 are supervised, the Bureau of Aeronautics may, when conducting service approval tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting service approval tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. The design and test data shall include the data obtained by the contractor in testing to Specification MIL-T-5422. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

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4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodially through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests.

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages and frequencies.

(3) Reliability test specified in 4.3.1.3, except that the test duration shall be 30 hours.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Bureau of Aeronautics.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

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(2) Vibration Test (90 minutes cycling - in each of 3 planes)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the Bureau of Aeronautics for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life tests shall be performed under the conditions specified in 4.4.1. The life test sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
26 to 200	1
201 to 500	1
Over 500	1 for each additional 500 or fraction thereof

4.4.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room temperature
Altitude	Normal room
Humidity	Room ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. Voltage	26.5 ± 2.5 volts

4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.4.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the Bureau of Aeronautics.

4.4.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

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(2) The circuit reference symbol-number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.4.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the Bureau of Aeronautics upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The Bureau of Aeronautics will review such proposals and determine whether they are acceptable.

4.4.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life tests shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the Bureau of Aeronautics.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics for review and approval. The right is reserved by the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. ECP's, EMC's and EMB's shall be prepared in accordance with SAR-300.

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5. PREPARATION FOR DELIVERY

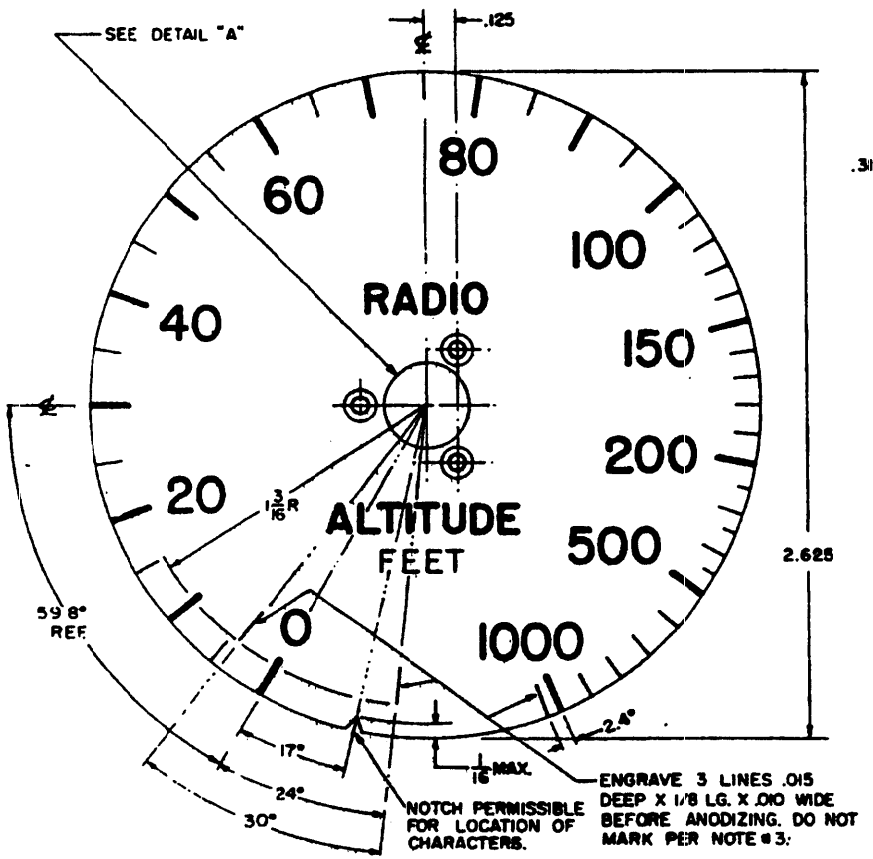
5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

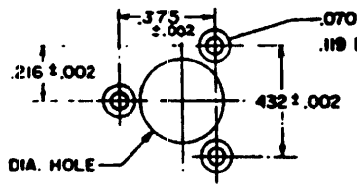
6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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DIAL

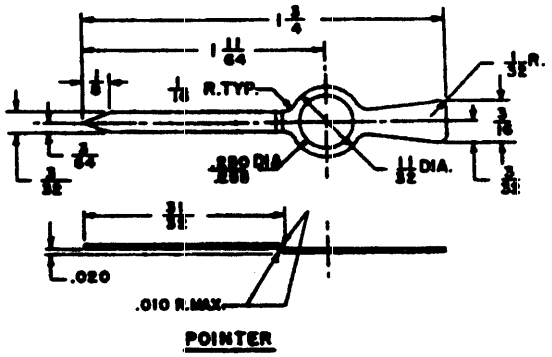


DETAIL "A"

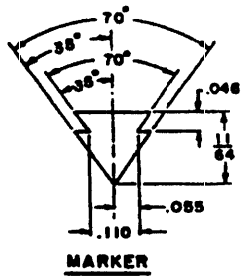
MARKING (REFERENCE)	HEIGHT OR LENGTH	WIDTH OF LINE
NUMERALS, MAJOR NUMERAL	.156	SEE NOTE 4
MAJOR GRAD.	.156	.030
MINOR GRAD.	.093	.015
SUBMINOR GRAD	.047	.015
LETTERING- RADIO	.125	SEE NOTE 4
ALTITUDE	.125	
FEET	.093	

CONCENTRICITY OF THE GRADUATIONS TO THE CENTER BORE SHALL BE HELD TO WITHIN .001 T.I.R.

LOCATIONS OF THE GRADUATIONS SHALL BE HELD TO WITHIN ±5 MIN.



POINTER



MARKER

ANGULAR LOCATION OF GRADUATIONS

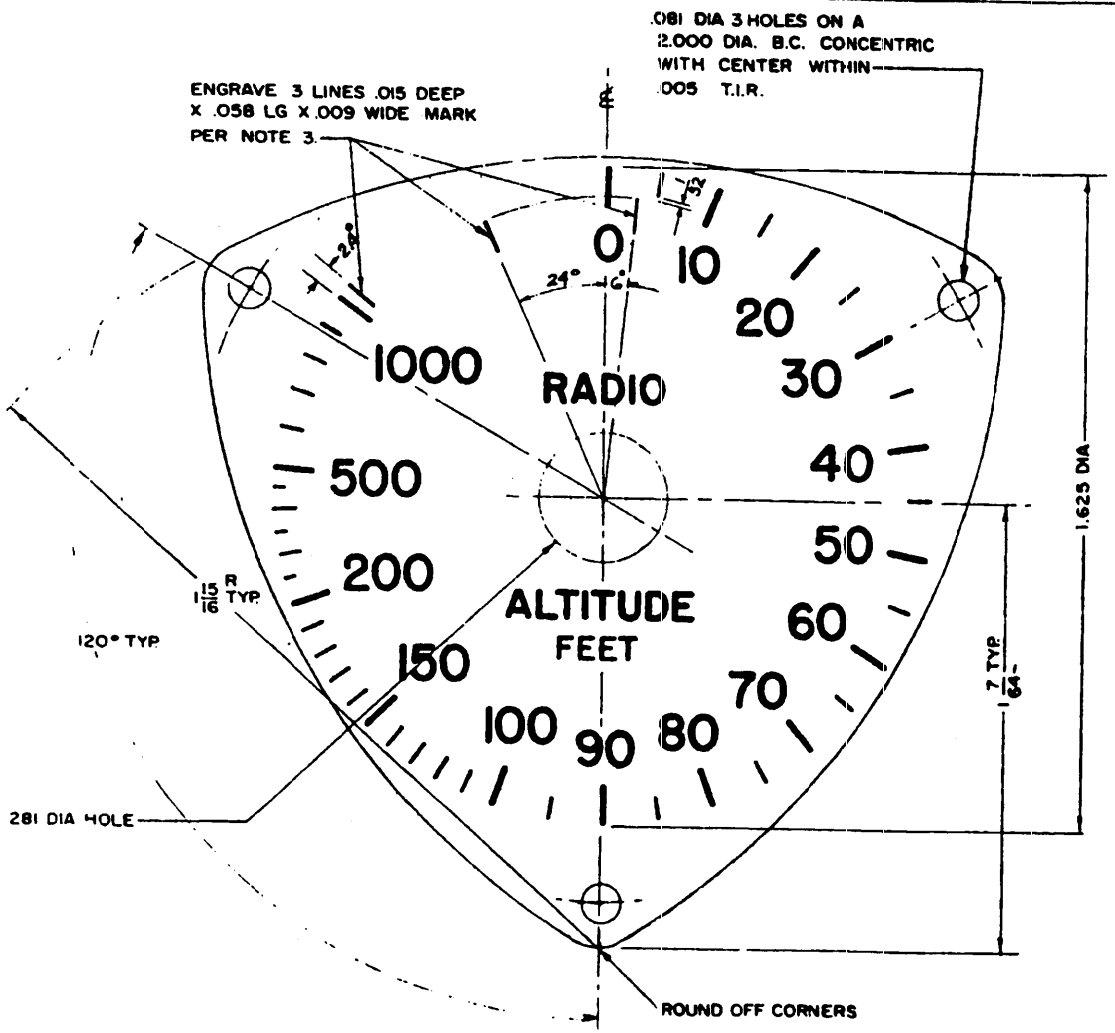
ALTITUDE	DEGREES	ALTITUDE	DEGREES
0	0	110	204.7
5	9.5	115	209.7
10	19.5	120	214.7
15	29.5	125	219.75
20	39.5	130	224.8
25	49.5	135	229.8
30	59.5	140	234.8
35	69.5	145	239.8
40	79.5	150	244.75
45	89.5	155	249.7
50	99.5	160	254.6
55	109.5	165	259.7
60	119.5	170	264.7
65	129.5	175	269.7
70	139.5	180	274.7
75	149.5	185	279.7
80	159.5	190	284.75
85	169.5	195	289.7
90	179.5	200	294.7
95	189.5	205	299.7
100	199.7	1000	304.7

NOTES:
1. SHALL BE FREE FROM BURRS AND SHARP EDGES.
2. BLACK ANODIZE PER QQ-Z-285 AND PAINT FRONT SURFACE WITH NONREFLECTING BLACK PAINT HAVING A COLOR CONFORMING WITH ANA BULLETIN NO. 157, COLOR 604 OR FED STD No. 595 COLOR 37038.
3. NUMERALS AND LETTERING TO BE IN ACCORDANCE WITH MS33558.
4. COLOR OF NUMERALS, POINTER, MARKER AND LETTERING TO BE WHITE IN ACCORDANCE WITH ANA BULLETIN No. 157 COLOR 601 OR FEDERAL STANDARD No. 595 COLOR 37875.

FIGURE 1.
HEIGHT CONTROL INDICATOR

MIL-R-21079(AER)

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MARKING (REFERENCE)	HEIGHT OF LENGTH	WIDTH OF LINE	WIDTH OF LETTER
NUMERALS:			
MAJOR NUMERAL	096	015	069
MAJOR GRAD	096	019	—
MINOR GRAD	058	009	—
SUBMINOR GRAD	029	009	—
LETTERING			
RADIO	.077	.015	056
ALTITUDE	.077	.015	.056
FEET	058	.012	.048

NOTE: CONCENTRICITY OF THE GRADUATIONS TO THE CENTER BORE SHALL BE HELD TO WITHIN .001 TIR LOCATIONS OF THE GRADUATIONS SHALL BE HELD WITHIN ± 8 MINUTES.

ANGULAR LOCATION OF GRADUATIONS (REF.)

ALTITUDE	DEGREES	ALTITUDE	DEGREES
0	0	110	204.7
5	9.5	120	209.7
10	19.8	130	214.7
15	29.6	140	219.75
20	39.8	150	224.8
25	49.8	160	229.8
30	59.8	170	234.9
40	69.8	180	239.9
40	79.8	190	244.95
45	89.9	200	250.0
50	99.9	250	254.9
55	109.9	300	259.9
60	119.9	350	262.9
65	129.9	400	266.5
70	139.9	450	270.8
75	149.8	500	274.55
80	159.7	600	281.7
85	169.5	700	288.4
90	179.35	800	294.3
95	189.1	900	300.6
100	198.8	1000	306.1

NOTES.
1. SHALL BE FREE FROM BURRS AND SHARP EDGES
2. BLACK ANODIZE PER QQ-Z-285 & PAINT FRONT SURFACE WITH NONREFLECTING BLACK ENAMEL PER 00-90348-1
3. MARK DIAL WITH FLORESCENT PAINT PER MIL-L-25142

FIGURE 2.
SERVO INDICATOR DIAL

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